ENTRIES 092-093 Beyond a Single Number — September 25, 2026 — Status: COMPLETE

Failure-Mode Breakdown, Weak-Link Identification, Chip Recommender

Three New Prediction Features, Built on Existing Validated Physics — No Retraining Needed

Why This Was Cheap

exact_dwell_cost (Entries 024-032) already computes success as a product of independent survival terms -- final-gate error, decoherence accumulated during SWAP dwell time, single-qubit gate cost, readout error -- it just never returned them separately, only the combined number. entry092_decomposed_prediction.py duplicates the SAME validated logic (same bug fixes, same closed-form derivations) but tracks each channel independently instead of folding them into one accumulator. Checked against the original exact_dwell_cost on 8 test pairs across two chips: every decomposed total matched the original to floating-point precision. This is exposition of physics already in the codebase, not a new model with new failure modes of its own to validate.

Entry 092a — Failure-Mode Breakdown

Given a circuit, reports not just "73% success" but which channel is costing the most: gate_error_survival, decoherence_survival, sq_gate_survival, readout_survival, ranked by loss. Example (Kyiv, qubits 77-82): total 60.4%, dominant failure mode decoherence (31.1 points of loss) vs gate error (0.6 points) -- on a long route, decoherence during the SWAP chain overwhelmingly outweighs the entangling gate's own error, exactly the kind of actionable distinction a single aggregate number can't offer.

Entry 092b — Weak-Link Identification

The per-SWAP-leg dwell log the original formula already builds (physical qubits, duration, raw gate error, dephasing) directly identifies the single worst leg by combined gate-error + dephasing score. For the same 77-82 example: weak link is edge (80,79), consistent with the decoherence-dominant verdict above. No new ground truth needed -- this was always computable from data the formula already touches, just never surfaced.

Entry 093 — Best Chip/Route Recommender

Turns the model from "grade this circuit" into "tell me where to run it." Built entirely on the existing 5-chip precomputed lookup (entry081) -- a ranking/filtering layer, not a new model. recommend(min_success, max_hop, max_uncertainty, chips, top_k) returns ranked candidates; the top result gets a live failure-mode breakdown attached automatically. Verified against three realistic queries: best 5 pairs with no filters (returned a mix of Kyiv and zero-shot Quebec candidates near 98%), best trained-chip pair under hop≤5 with explanation, and a zero-shot-only query ("chip I haven't tried yet") -- all returned sane, ranked results.

Shipped

Both new capabilities are live in the API, not just standalone scripts: POST /predict/breakdown and GET /recommend, tested end-to-end against a running server (both returned correct 200 responses with the expected structure). README updated with usage examples.

Entries 092-093 — Conclusion

All three requested prediction features (failure-mode breakdown, weak-link ID, chip recommender) are built, validated, and wired into the live API in one pass -- made possible because the physics-informed v4.1 formula already carried this information internally from Entries 024-032, it just needed to be exposed rather than re-derived. No new training data or model retraining was required for any of the three. Next up per the agreed plan: multi-bell-pair template scaling (Entries 089-091's track).

Research Log — Index Addendum (Entries 092-093)

Files: entry092_decomposed_prediction.py, entry093_recommend.py, api/main.py (/predict/breakdown, /recommend), api/README.md.